UNIVERSITY OF CAMBRIDGE

It is all about data!

Focus of the day: It is all about data!

- Introduction to financial data
- How to transform financial data
- Perform Exploratory Data Analysis (EDA)
- Tool-kit overview
- Data access
- Digging a little deeper

BIG DATA & AI

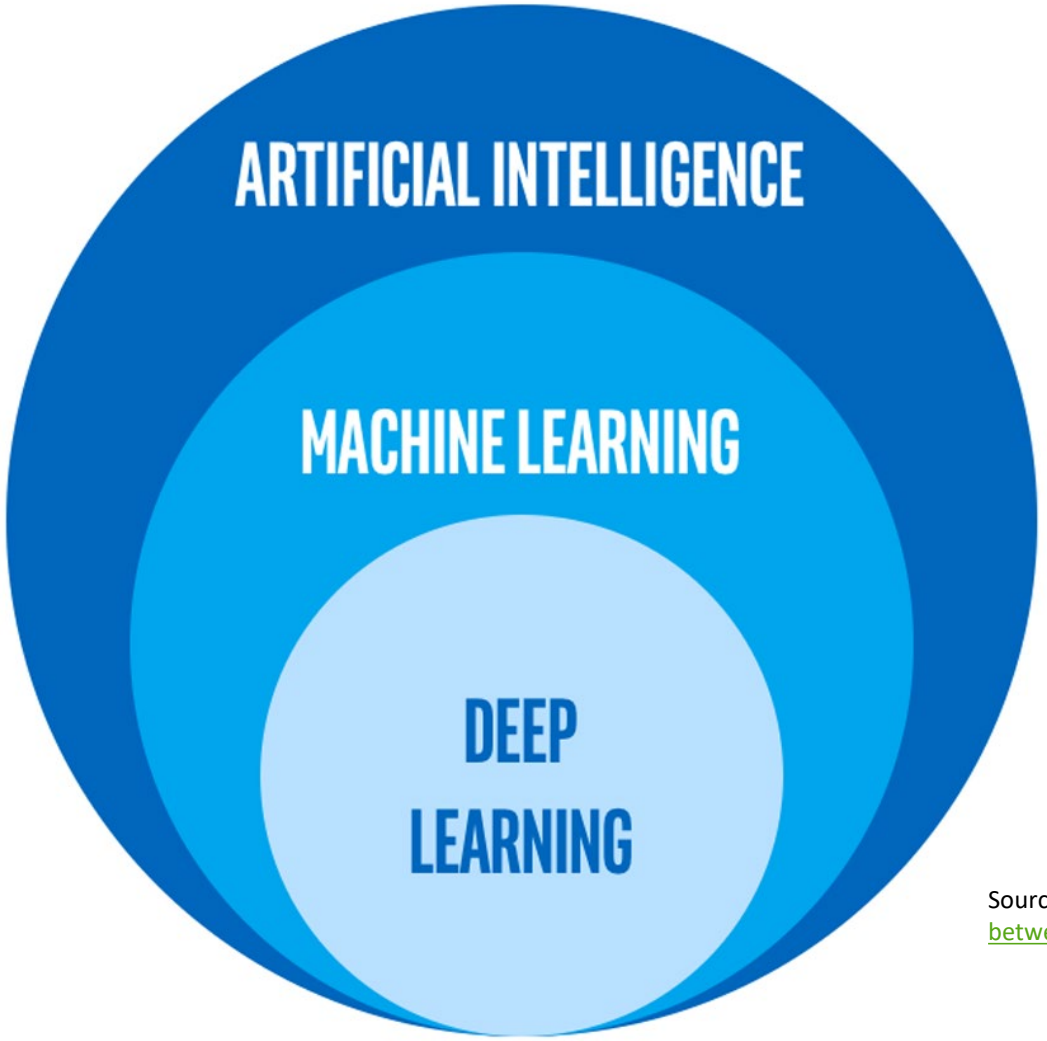
AI as a discipline has been around for a long time, why now?

An Early AI Experiment…..

Claude Shannon demonstrates "Theseus" Machine Learning @ Bell Labs (High Quality)

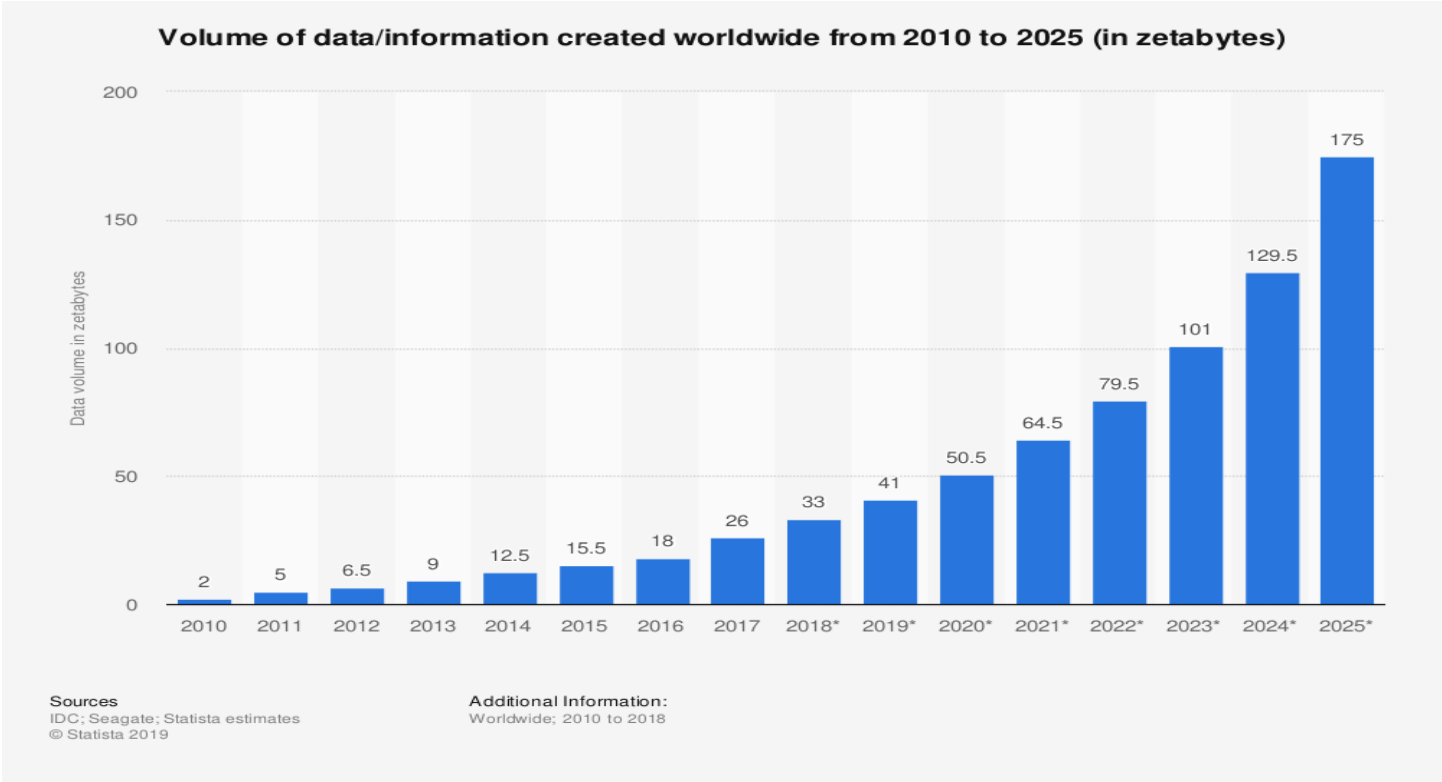
https://youtu.be/_9_AEVQ_p74

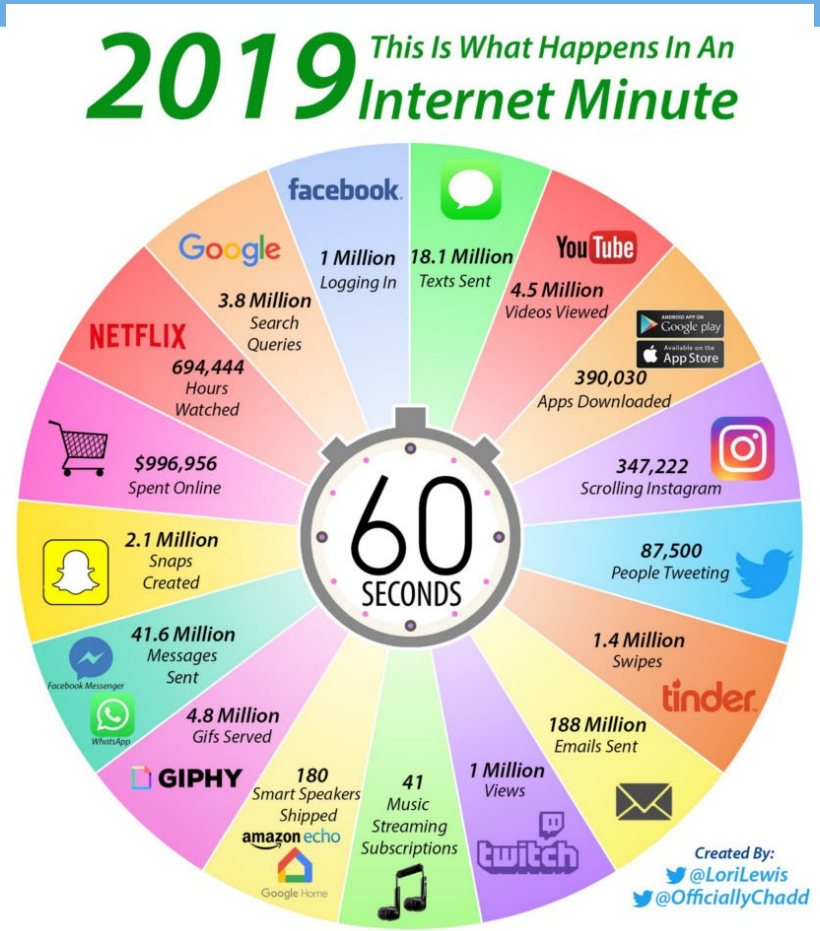
A bit of terminology first



Source: https://www.intel.com/content/www/us/en/artificial-intelligence/posts/difference-between-ai-machine-learning-deep-learning.html

The Data “Revolution”





Source: Infographic: What Happens in an Internet Minute in 2019? (visualcapitalist.com)

Data “Accounting”

Unit	Value	Size
bit (b)	0 or 1	1/8 of a byte
byte (B)	8 bits	1 byte
kilobyte (KB)	1000^1 bytes	1,000 bytes
megabyte (MB)	1000^2 bytes	1,000,000 bytes
gigabyte (GB)	1000^3 bytes	1,000,000,000 bytes
terabyte (TB)	1000^4 bytes	1,000,000,000,000 bytes
petabyte (PB)	1000^5 bytes	1,000,000,000,000,000 bytes
exabyte (EB)	1000^6 bytes	1,000,000,000,000,000,000 bytes
zettabyte (ZB)	1000^7 bytes	1,000,000,000,000,000,000,000 bytes
yottabyte (YB)	1000^8 bytes	1,000,000,000,000,000,000,000,000 bytes

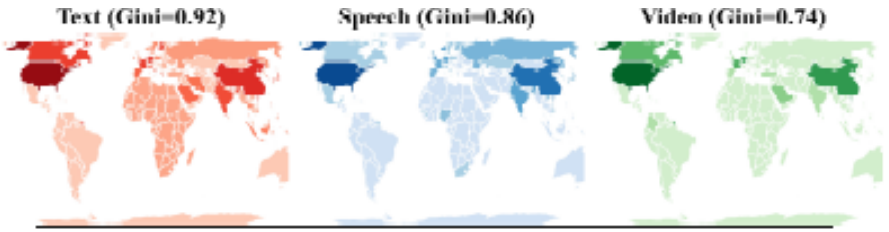
MEASURE STORAGE – TECHNOFLOWGY (WORDPRESS.COM)

DATA TRANSPARENCY, BIASES AND ACCESS

- Data is the fuel that powers AI systems
- Understanding the data that is used to develop and train AI is key
- Be aware of hidden biases that may impact results
- Don’t take access for granted

DATA TRANSPARENCY, BIASES AND ACCESS

- Rising importance of web, social media and synthetic sources
- Language and geographical distribution of training data sources
- Potential restrictions at source or platform-level



	Africa	Asia	Europe	N America	Oceania	S America
By Count						
Text	0.3	13.4	24.0	61.5	0.7	0.2
Speech	3.6	35.7	30.4	30.4	0.0	0.0
Video	0.0	25.2	24.4	48.0	0.8	1.6
By Tokens or Hours						
Text	0.0	6.1	55.4	38.4	0.1	0.0
Speech	0.1	38.8	18.8	42.4	0.0	0.0
Video	0.0	23.1	22.0	38.2	16.7	0.1

Figure 3: The geographical distribution of countries (world maps) and continents (table) represented by dataset creators. **Despite some differences in European, Russian, and Middle Eastern representation, creators are heavily concentrated in the US, China, and Western Europe, with little to no representation in South America or Africa, across modalities.** The current Gini coefficient for (Text, Speech, Video) = (0.92, 0.86, 0.74), where higher values indicate more concentration.

Source: The Data Provenance Initiative, 2024

TYPE OF FINANCIAL DATA

Type of data

- There are fundamentally four types of data:
  - Fundamental data
  - Market Data
  - Alternative data
  - Analytics

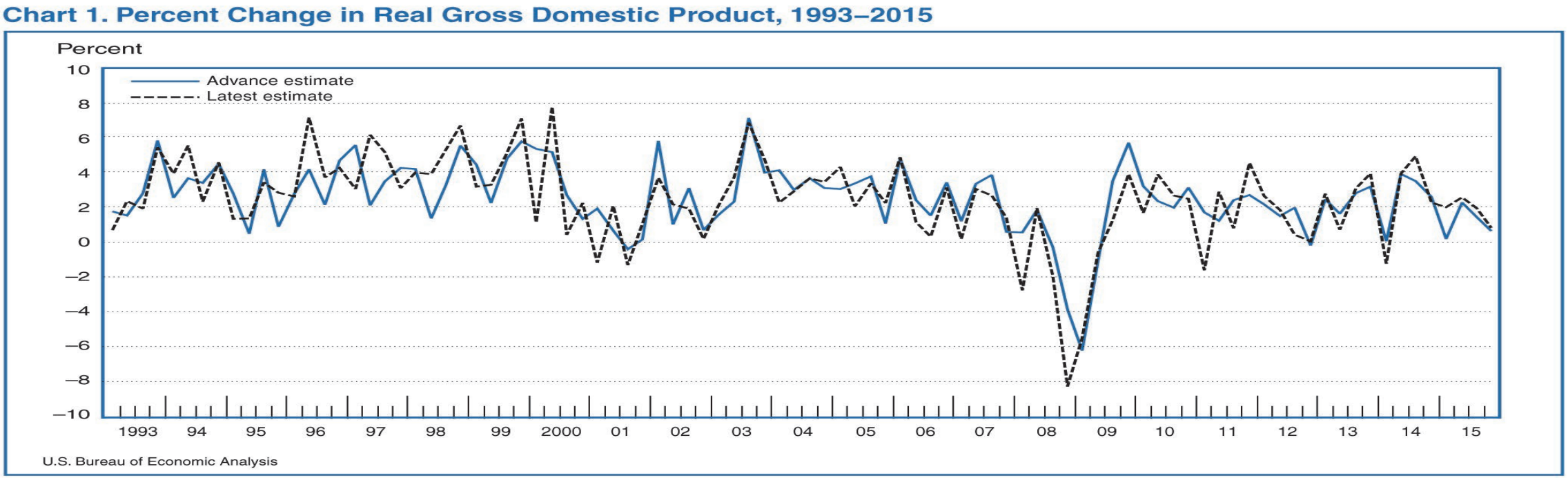
Type of data-Fundamental data

- Info about the macro-economy published by official statistical offices; Info about a company's business performance and it can be found in regulatory filings and business analytics.
- It includes:
  - For economy: GDP, inflation, trade volumes, confidence indicators
  - For companies: Assets, liabilities, sales/revenue, costs, earnings
- Usually low frequency and published with a lag
- Subject to backfilling/re-stating

Focus-Fundamental Data & Backfilling/Restating

- Let's look at US GDP data
- The BEA quarterly releases: *advanced* estimates (a month after quarter end); *second* estimate (two months after quarter end); *third* estimate (3 months after quarter).Each is known as vintage. In addition, there are four vintages of annual estimates-early annual, first annual, second annual, third annual. Plus there are comprehensive NIPA reviews that happen about every 5 yrs.
- Reasons for revisions include missing data or methodological changes
- Why does it matter?

Focus-Fundamental Data & Backfilling/Restating



Source: The Revisions to GDP, GDI and their Major Components by Dennis J Fixler et al , 2018

Type of data-Market data

- Data related to trading activity and includes:
  - Price, volume, implied volatility etc
- Tends to be high frequency
- More challenging to process
- Could be costly

Type of data-alternative data

- Alternative data in finance is data that is not yet reflected in the fundamental and/or market data but may be a leading indicators for them . It may include
  - Data produced by individuals (social media, web searches), businesses (e.g. credit card transactions),and sensors (e.g. satellite data, CCTV) - *source: Kalnovic and Krishnamachari [2017]*
- Potential for significant information advantage,but could be hard to process and costly to acquire.

Type of data-analytics data

- Derived data from either fundamental or market data or alternative; its value (i.e. info advantage) is not the original info per se, but how it has been processed. It includes:
  - Proprietary sentiment indicators, economic now-casting
- Inherently less transparent and less readily available

FINANCIAL DATA TRANSFORMATION

Financial data transformation

- Normalizing the data by subtracting the mean and dividing by standard deviation
- *Take the log of the data*- this has a number of advantages in financial application including additivity and symmetry with respect to gains and losses. Very common when dealing with price data
- Bucket into evenly distributed buckets and create a categorical variable from that
- Turn the data into binary variable based on some threshold value with data =1 if above threshold else zero.

EXPLORATORY DATA ANALYSIS

EDA: Basic steps

- Any financial modelling should start with an understanding of the data you have
- Plot over different frequencies
- Identify seasonality effects
- Plot using transformation techniques discussed earlier

EDA: Analysing price data

- Most work in financial modelling has to do with price data so it is key to understand key techniques to analyse time series of prices:
  - Fixed time intervals("time bars"): most common way of accessing/displaying financial time series e.g. daily price of a stock at market close; challenge of this approach is that it treats all intervals the same: is the info value of a stock price on the 24/12 the same as 24/11?
  - Tick bars: sample based on a predefined number of transactions; helps capture info arrival; be aware of impact of opening and closing auction; closer to an IID Gaussian distribution.
  - Volume bars: sample intervals based on a pre-defined amount of security's units being exchanged; it may help address biases we see in tick bars due to order fragmentation. Better statistical properties i.e. closer to an IID Gaussian distribution.
  - Dollar bars: sample intervals based on a predefined amount of market value being exchanged (of course, in any currency!); addresses biases due to changing number of shares; may need to adjust sample bar size through time.
  - Information driven bars: sample as a function of new info arrival/"informed actors"- linked to the complex literature around market microstructure theories.

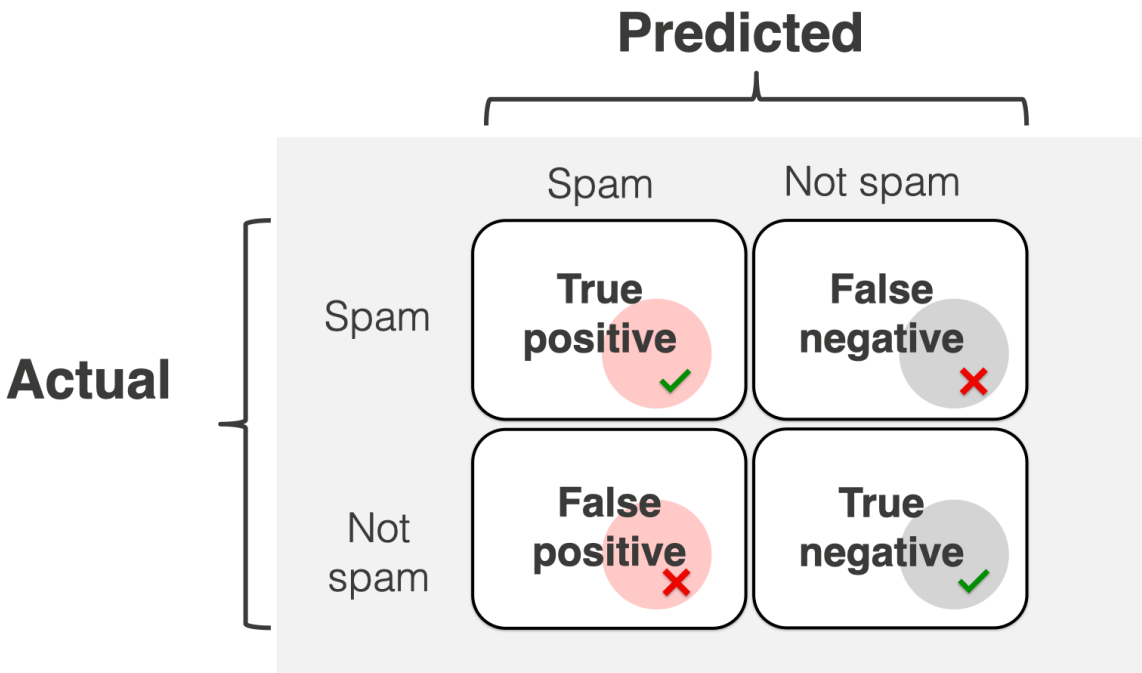
Focus- “Time bars”

- Time bars are obtained by sampling information at fixed time intervals; it usually includes:
  - Time-stamp
  - Volume weighted average price (VWAP)
  - Open (i.e. first) price
  - Close(i.e. last) price
  - High price
  - Low price
  - Volume traded

STATISTICS REFRESHER

Using the **confusion matrix**, you can visualize all 4 different outcomes in a single table.

- Accuracy=(TP+TN)/(TP+TN+FP+FN)
- Precision=TP/(TP+FP)
- Recall=TP/(TP+FN)



EXERCISE

ACCESSING EARNINGS CALL TRANSCRIPTS

- Using either Bloomberg or LSEG based products, access the earnings call transcripts for CY 2024 for the companies in the DAX
- Please note index membership changes
- Use bulk download functions when available

DIGGING A LITTLE DEEPER

DIGGING A LITTLE DEEPER

- To dig a little deeper in the concepts covered in this lecture, below are a few additional optional readings:
  - Market micro-structure: Chapter 19 of Advances in Financial Machine Learning by De Prado
  - Time Stamps and Financial Modelling: Chapter 6 of Doing Data Science by Cathy O'Neil & Rachel Schutt
  - Revisions to economic data: excellent examples can be found in https://obr.uk/box/rewriting-recent-history-real-gdp-revisions/; https://apps.bea.gov/scb/pdf/2018/01-January/0118-revisions-to-gdp-gdi-and-their-major-components.pdf

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